

Lesson Plan Generator

Feature Scope & Specification · TEAM-aligned · deterministic · print-first

Product: U.S. History Hack™ (TroopToTeacher Technologies, LLC) **Pilot:** Franklin High School, Williamson County Schools **Status:** Approved scope — ready for build

1 · Goal

Let a teacher click “*Generate Lesson Plan*,” choose the activities/organizers and the class-period length, and receive a 100% TEAM-aligned, printable lesson plan plus a single ZIP of every material needed for the day — assembled deterministically from existing U.S. History Hack content. Everything is print-first.

2 · Non-negotiable principle — deterministic assembly, no runtime AI

The generator **composes** a plan from pre-authored, human-reviewed building blocks against a fixed TEAM template. It **must not** call a language model at runtime. This preserves the certified compliance posture (“no AI at runtime; no student data to any AI”). The teacher’s picks drive a rule-based assembler — the same deterministic pattern as the existing adaptive/spaced-repetition logic.

3 · What we reuse (already built)

Existing piece	Role in this feature
lib/lesson-package/ (builder, types, manifest, download)	The deterministic assembly engine + picker UI + /print route + URL round-trip. Generalize beyond Unit 1.
pdf-lib + jszip (client-side)	Merge print-ready PDFs and bundle the one ZIP, in-browser (static-export safe).
Per-standard content (I-CAN, DOK banks, tiered + ELL/IEP worksheets, CER/HIPP, primary sources)	The pickable building blocks; already keyed to US.01–US.95.
Feature-flag + role gating	Ship behind a teacher-role gate and an optional NEXT_PUBLIC flag.

4 · Teacher flow

1. Open “Generate Lesson Plan”; pick the standard (US.NN) and date.
2. Pick the schedule (defaults to the school’s bell schedule; auto-detects late-start dates) — or custom/block.
3. Pick activities & graphic organizers; the app recommends a set for the standard (rule-based).
4. Review the live preview (TEAM sections auto-fill; teacher adds grouping/feedback + environment notes).
5. Generate → TEAM-aligned lesson plan (print) + one ZIP of the day’s materials + a “where to print” sheet.

5 · TEAM-aligned plan template

Sections map to the TEAM General Educator Rubric (3 domains, 19 indicators). **Auto** = fills from picks + existing data; **Prompt** = teacher input.

Plan section	Source	Notes
Standards & Objectives	Auto	I-CAN + TN standard text + sequenced sub-objectives + prior learning
Assessment & Evidence	Auto	formative + form A/B + CER/HIPP + success criteria (TEAM Planning: Assessment)
Lesson Structure & Pacing	Auto	timed segment table for the chosen schedule
Activities & Materials	Auto	the teacher’s picks
Questioning	Auto	DOK 1–4 ladder from the item bank, text-based
Thinking / Problem-Solving	Auto	tagged from the chosen activities
Grouping & Feedback	Prompt	teacher selects arrangement + feedback checkpoints
Differentiation	Auto	ELL/WIDA + IEP/504 from tiered materials
Reflection / Extension / Reteach	Prompt+Auto	post-lesson reflection; enrichment; one-click remediation path

Plan section	Source	Notes
Environment reminders	Prompt	climate/norms checklist (observed, not generated)
Materials & Where to Print	Auto	ZIP contents + deck pointer
Appendix: TEAM indicator map	Auto	each indicator → where addressed (toggle)

Honest claim: earns a defensible 5 on the Planning domain (what a written plan is scored on) and documents every Instruction indicator; the Environment domain is prompted, not guaranteed.

6 · Schedule & pacing — configurable per district & per school

Requirement: bell schedules, calendars, and pacing are **NOT hardcoded**. They are editable settings owned by each district and, where schools differ, each school. FHS's 47/43/41 is **seed data, not a constant**. Values cascade District → School → Teacher; each level can override the one above.

Level	Owns / sets	Editable?
District	Default bell schedules, instructional calendar (start/end, closures, grading periods), late-start dates, default unit/standard pacing	Yes — district admin
School	Override bell schedules & late-start dates when the school differs (e.g., FHS 47/43/41 vs. another WCS high school)	Yes — school admin
Teacher	Per-lesson: pick a schedule (Regular/Activities/Late-Start), enter custom minutes (e.g., 50), or block (e.g., 90); nudge their own pacing dates	Yes — per lesson

Defaults & behavior

- Schedule picker defaults to the teacher's school schedule; auto-selects the short (late-start) schedule on the school's configured late-start dates.
- Pacing defaults to auto-distributing units/standards across the district's real instructional days; teacher can override (compress/expand, move a date).
- Seed the pilot with FHS (47 Regular / 43 Activities / 41 Late-Start, 27 late-start dates) + the WCS 2026–27 calendar — as editable data, not code. (App note: current min-46 → min-47 for FHS Regular.)

7 · Per-standard materials (the content-granularity fix)

Workbooks are authored by unit; a teaching day is by standard. Target: split at the source so the ZIP carries just today's standard.

- A — Split at source (target):** extend the build scripts to emit per-standard sections (Unit 1 already ships this way: us-01...us-07). Biggest lift, but script extension over data you already have.
- B — Page-range slice (bridge):** store a per-standard page map and cut ranges with pdf-lib at ZIP time for units not yet split.
- C — Point to pages (fallback):** plan says "print pages X–Y." Zero build cost; the floor so nothing blocks launch.

8 · Decks — download-once resource, not bundled

Decks are unit-level and reused all week, and they're being updated, so they're **pointed to, not zipped**: the plan names the deck + the standard's slide range and links to Lecture Decks (a pointer always resolves to the current deck; a bundled copy goes stale). Teacher decks stay behind their existing secure gate.

9 · Remediation — a preset of the generator

From the mastery data, a "Remediate US.NN" button runs the same assembler with intervention defaults (entry-tier worksheet, targeted DOK 1–2, ELL/IEP supports) → a reteach ZIP + short reteach plan. Rule-based (no AI). This is the MTSS/RTI² Tier 2–3 story; benchmark windows (per the district assessment calendar) are the natural triggers. Trigger needs live mastery data; resources work regardless.

10 · The daily ZIP

One role-gated (teacher) download. Suggested layout:

Folder	Contents
00_START_HERE	TEAM lesson plan + “Where to Print” checklist
1_PRINT_student	student workbook pages, worksheets, organizers (PDF)
2_PRINT_teacher	teacher guide pages, answer keys (PDF)
3_EDITABLE	same items as DOCX
(decks)	NOT included — linked as a unit resource

- **Student-only toggle:** omit teacher folders for a safe hand-out.
- Only real files are bundled (honest availability); size stays modest since decks are excluded.

11 · Naming, copyright & trademark

- Every generated artifact carries: footer + file metadata — © 2026 TroopToTeacher Technologies, LLC · U.S. History Hack™ · All rights reserved.
- Use ™ (common-law), not ® (not federally registered yet). Copyright notice is valid now; registration is a recommended to-do, not a blocker.
- Product family: U.S. History Hack / World History Hack / Government Hack. Default everything to U.S. History Hack unless working on the others.

12 · Build phasing & effort

Phase	Scope	Relative effort
Pilot	TEAM template on the existing engine; 1 unit (e.g., Unit 1, already split); FHS schedule seeded; ZIP + remediation preset	Small (code); days
Config layer	District/School/Teacher schedule + pacing settings (editable, cascading)	Medium
Content scale-out	Per-standard split + duration metadata for units 2–10; index printables into the manifest	Largest — content pipeline, not code
Verify	TEAM template checked against the official rubric; per-standard QA	Medium

13 · What’s needed / open items

- Per-activity duration metadata (durationMinutes) so time fills from picks — the single most load-bearing missing field.
- Bell schedules for any school beyond FHS (each school’s own PDF).
- Per-standard split + manifest indexing for units 2–10 (decks currently exist only for Units 1, 5, 10).
- TEAM template verified against the official current General Educator Rubric.
- Trademark registration (in motion) — keep ™ until granted.